

ECON*6140 ECONOMETRICS I

Assignment #1

Fall 2026.

Due: Wednesday, October 14, 2026.

(1)

Suppose we have the linear model $y = X\beta + u$, where X is $n \times k$, β is $k \times 1$ and y and u are $n \times 1$. We partition $X = [X_1|X_2]$ and $\beta = (\beta_1 \ \beta_2)^T$, where X_1 is $n \times k_1$, X_2 is $n \times k_2$, β_1 is $k_1 \times 1$ and β_2 is $k_2 \times 1$, $k = k_1 + k_2$. Hence

$$y = [X_1|X_2] \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} + u = X_1\beta_1 + X_2\beta_2 + u.$$

Using the formula for the OLS estimator $b = (X^T X)^{-1} X^T y$ to find expressions in terms of the partitions X_1 , X_2 for b_1 and b_2 . In other words solve

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \left[\begin{pmatrix} X_1^T \\ X_2^T \end{pmatrix} (X_1 \ X_2) \right]^{-1} \begin{pmatrix} X_1^T \\ X_2^T \end{pmatrix} y.$$

Hint: Use the formula for the inverses of partitioned matrices in Johnston and Dinardo pp. 472–473. The resulting b_1 , b_2 should be expressed in terms of X_1 , X_2 , M_1 and M_2 .

(2)

The file `tbrate.docx` contains data for 1950:1 to 1996:4 for three series: r_t the interest rate on 90-day treasury bills, π_t the rate of inflation, and y_t the logarithm of real GDP. For the period 1950:4 to 1996:4 run the regression

$$\Delta r_t = \beta_1 + \beta_2 \pi_{t-1} + \beta_3 \Delta y_{t-1} + \beta_4 \Delta r_{t-1} + \beta_5 \Delta r_{t-2} + u_t \quad (1)$$

where Δ is the first difference operator, defined so that $\Delta r_t = r_t - r_{t-1}$. Plot the residuals and fitted values against time. Then regress the residuals on the fitted values and on a constant. What do you notice? Now regress the fitted values on the residuals and on a constant. What do you notice in this last regression?

(3)

For the same sample period regress Δr_t on a constant, Δy_{t-1} , Δr_{t-1} and Δr_{t-2} . Save the residuals from this regression and call them e_t . Then regress π_{t-1} on a constant, Δy_{t-1} , Δr_{t-1} and Δr_{t-2} , save the residuals from this regression and call them e_t^π . Now regress e_t on e_t^π . How are the estimated coefficient and the residuals from this last regression related to anything that you obtained when you estimated regression (1) in question (2) above?

(4)

Using the data from the consumption file to be found at the course link page, construct the variables c_t , the logarithm of consumption and y_t the logarithm of income and their difference $\Delta c_t = c_t - c_{t-1}$ and $\Delta y_t = y_t - y_{t-1}$. Use the data to estimate the following model for the period 1953:1 to 1996:4

$$\Delta c_t = \beta_1 + \beta_2 \Delta y_t + \beta_3 \Delta y_{t-1} + \beta_4 \Delta y_{t-2} + \beta_5 \Delta y_{t-3} + \beta_6 \Delta y_{t-4} + \text{errors} \quad (2)$$

Let $\gamma = \sum_{i=2}^6 \beta_i$. Calculate $\hat{\gamma}$ and its standard error in two different ways. In one method use the result where if $\hat{\gamma} = w^T \hat{\beta}$ where w is a vector of known coefficients then $\text{Var}(\hat{\gamma}) = w^T \text{Var}(\hat{\beta}) w$. The other method should use a transformation of the equation (1) above which allows $\hat{\gamma}$ and its standard error to be read directly from the regression output.

(5)

Consider the following linear regression

$$y = X_1 \beta_1 + X_2 \beta_2 + u$$

where y and u are $n \times 1$, X_1 is $n \times k_1$, X_2 is $n \times k_2$, β_1 is $k_1 \times 1$ and β_2 is $k_2 \times 1$. Let $\hat{\beta}_1$ and $\hat{\beta}_2$ be the OLS parameter estimates from running this regression.

Now consider the following regressions all to be estimated by OLS:

- (a) $y = X_2 \beta_2 + u$
- (b) $P_1 y = X_2 \beta_2 + u$
- (c) $P_1 y = P_1 X_2 \beta_2 + u$
- (d) $P_X y = X_1 \beta_1 + X_2 \beta_2 + u$
- (e) $P_X y = X_2 \beta_2 + u$
- (f) $M_1 y = X_2 \beta_2 + u$
- (g) $M_1 y = M_1 X_2 \beta_2 + u$
- (h) $M_1 y = X_1 \beta_1 + M_1 X_2 \beta_2 + u$
- (i) $M_1 y = M_1 X_1 \beta_1 + M_1 X_2 \beta_2 + u$
- (j) $P_X y = M_1 X_2 \beta_2 + u$

Here P_1 projects orthogonally on to the space of the X_1 , $M_1 = I - P_1$ and P_X projects orthogonally on to the space of all the X 's, $M_X = I - P_X$. For which of the above regressions are the estimates of β_2 the same as in the original regression and why? For which are the residuals the same and why?

Transcription notes

- The original is three photographed pages of the printed assignment sheet (the photographs are landscape, with the page lying on its side). Page 1 carries the letterhead and questions (1) and (2) down to “Then regress the residuals”; page 2 carries the rest of (2), then (3), (4) and (5) down to item (a); page 3 carries items (d)–(j) and the closing paragraph of (5).
- The header block is reproduced from the sheet: Department of Economics and Finance, Lang School of Business and Economics, ECON*6140 Econometrics I, Instructor: T. Stengos, Fall 2026, Assignment #1, due Wednesday, October 14, 2026. The instructor line is not repeated in the title block above.

- The two numbered equations keep the numbers the sheet gives them: the dynamic regression of Δr_t in (2) is tagged (1), and the distributed-lag model for Δc_t in (4) is tagged (2), so the tags run in order of appearance on the sheet, not in question order.
- In question (4) the sheet says “a transformation of the equation (1) above” although the equation immediately above it is tagged (2). The wording is kept as printed; the reference is evidently to equation (2), which is the model of that question.
- Typographical slips on the sheet, corrected here: “teh residuals” for “the residuals” in (2); “Npw regress” for “Now regress” in (3); “transfomration” and “standrad error” for “transformation” and “standard error” in (4); “course course link page” for “course link page” in (4), where the first “course” ends a line and the second begins the next; “the space of tall the X ’s” for “all the X ’s” in (5).
- The sheet writes the transpose with a capital T superscript throughout — $X^T X$, $(\beta_1 \beta_2)^T$, X_1^T — where these notes usually use the prime mark $'$. Kept as printed.
- Equation (2) names its disturbance “errors” rather than u , and the regressions of (5) all write “+ u ” with the same symbol for the disturbance as the original model; both kept as printed.
- The ten items of question (5) are lettered (a)–(j) on the sheet, the first three on page 2 and the remaining seven on page 3; they are typeset with the same letters (a)–(j) here, in the same order.
- Nothing in the three photographs was illegible. Read with the matrices as P_1 , $M_1 = I - P_1$ for the space of X_1 and P_X , $M_X = I - P_X$ for the space of all the X ’s, as the paragraph before the question states.